

Particle-Resolved Direct Numerical Simulation of Turbulence-Cloud-Aerosol Interactions

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Clouds play a critical role in the evolution of weather and climate. They consist of numerous tiny water droplets that grow as water deposits on the background aerosol particles. The objective of the presented work is to better understand the aerosol-cloud interactions. Previous developments such as global climate models, numerical weather prediction, and large eddy simulation (LES), could not represent some of the important processes that occur at spatial scales smaller than the typical grid sizes used in these models, including the microphysics of clouds and aerosols, and the turbulent mixing between cloud droplets and environmental air. We adopt a particle-resolved direct numerical simulation (PR-DNS) approach and aim to resolve the smallest eddies in the flow, track the cloud and aerosol particles individually, and cover a domain size of practical interest. We monitor the activation of aerosol particles into cloud droplets and the deactivation of cloud droplets into aerosol particles by varying turbulence intensity and dry aerosol size distribution. A detailed analysis of atmospheric turbulence in our model is also presented. Our results indicate that dry aerosol size distribution influences the activation process. When particles are smaller, the curvature and solute effects are more pronounced. We also find that turbulence enhances the deactivation of cloud droplets.

I. Nomenclature

$\mathbf{u}$	=	instantaneous velocities in the x, y, z coordinate directions
$\mathbf{F}_b, \mathbf{F}_c$	=	buoyancy term, external forcing term
T, q_v	=	instantaneous temperature, instantaneous water vapor mixing ratio
C_d	=	condensation rate
S_e, S, S_k	=	environmental supersaturation, particle equilibrium saturation ratio, equilibrium supersaturation
ϵ	=	dissipation rate
$\mathbf{X}, \mathbf{V}$	=	droplet position vector, droplet velocity vector
τ_p	=	particle response time
η, ν	=	Kolmogorov length scale, kinematic viscosity
ρ_l, ρ_a	=	density of water, air
$E(k)$	=	kinetic energy spectrum
μ_T, μ_v	=	molecular diffusivities for temperature, water vapor mixing ratio
$R_{ij}, \mathbf{r}$	=	correlation coefficient, separation distance

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II. Introduction

In the atmosphere, aerosol particles grow in a supersaturated environment and activate into cloud droplets. The opposite happens or cloud droplets deactivate into aerosol particles in a subsaturated environment. Our goal is to understand the activation and deactivation processes through numerical investigation. Since experimental measurements are not plenty and always feasible, computational models can be a good alternative to produce data of our interest. But numerical representation of clouds is still challenging. We review some literature on the modeling of turbulent clouds in the next few paragraphs.

Kumar et al. [1] studied turbulent mixing and entrainment at the boundary of a cloud using a DNS model that couples the Eulerian description of turbulent velocity, temperature, and water vapor mixing ratio fields with a Lagrangian ensemble of cloud droplets that grow and shrink by condensation and evaporation, respectively. Lanotte et al. [2] combined the Eulerian method for turbulent velocity field with the Lagrangian method for cloud droplets. They discussed the role of turbulence on droplet condensation at moderate to high Reynolds numbers. Vaillancourt et al. [3] solved the incompressible Navier-Stokes equations by DNS approach and tracked the droplets individually. It was suggested that the finite inertia of particles in a turbulent field causes particles to diverge from high vorticity regions and to converge preferentially in the regions of low vorticity. Gao et al. [4] applied a particle-resolved direct numerical simulation (PR-DNS) model and simulated three initial configurations for decaying and forced turbulences, each. Here, initial configuration refers to the arrangement of cloudy and cloud-free regions in the computational domain at the beginning.

The issue of turbulence forcing is important when simulating atmospheric turbulence in numerical models. Yu Li et al. [5] conducted a PR-DNS study to the effects of turbulence on temperature, water vapor mixing ratio, supersaturation, and buoyancy force. They compared their results for three different Kolmogorov scales- 0.54 mm, 0.65 mm, and 0.89 mm, for different Taylor scale Reynolds numbers and dissipation rates. Gao et al. [4] implemented homogeneous isotropic turbulence (HIT) in the context of cloud microphysics by initializing the velocity field in the Fourier space. Eswaran et al. [6] demonstrated that spectral forcing can lead to statistically stationary velocity fields in DNS studies of homogeneous, isotropic flows. Rogallo et al. [7] devised a spectral forcing scheme and computed the correlation coefficients.

We review the literature on the role of aerosol particles during cloud droplet growth. Petters et al. [8] showed that a single parameter, namely kappa (κ), can represent hygroscopic growth. They improved the formulation of the Kohler theory by introducing kappa to compute critical supersaturation. Chen et al. [9] discussed the altitude dependency of aerosol-cloud interactions. They found that above a maximum supersaturation, some cloud droplets deactivated into aerosol particles. More investigation by Chen et al. [10] suggested that presence of aerosol particles led to more but smaller cloud droplets. Rothenberg [11] investigated the influence of physical and chemical properties of aerosols on the activation process in detail.

We also review the literature on turbulence-particle interactions in the atmosphere. Shaw [12] discussed the turbulence-cloud interactions at small scales. The author commented that cloud droplets have inertia, and interact with their environment thermodynamically, leading to processes such as buoyancy reversal that tend to enhance intermittency. By intermittency, it is meant that the probability of large-amplitude fluctuations is greatly increased relative to what might be expected for a normally distributed random variable. Shawon et al. [13] examined the dependence of aerosol-cloud interactions on turbulence in a laboratory setting. They identified that turbulence causes more aerosol particles to be activated, and the activated fraction decreases monotonically as the concentration of aerosol increases. Ditas et al. [14] also measured supersaturation fluctuations in a turbulent field. They interpreted that supersaturation fluctuations depend on the combined fluctuations of temperature and water vapor mixing ratio.

Despite the recent progresses, a knowledge gap exists in our understanding of turbulence-cloud-aerosol interactions. Hence, we modify the PR-DNS model developed by Gao et al. [4] and incorporate the kappa-Kohler theory to study both activation and deactivation. The direct numerical simulation (DNS) approach [15] resolves all spatial scales and produces most accurate results without requiring sub-grid scale parametrizations. But it comes with a high computational cost. The Lagrangian approach to track individual particles [16] is free of numerical diffusion unlike the Eulerian approach which follows the evolution of particle number concentration. The former is more accurate but computationally expensive. To produce our results, we vary turbulence intensity, initial supersaturation, and dry aerosol size distribution. We report on the thermodynamic properties, and particle size distributions of aerosol particles and cloud droplets. We also explain the influence of turbulent fluctuations on cloud and aerosol microphysics.

III. Governing Equations

The dynamic field

The turbulent air is governed by the Navier-Stokes equations with Boussinesq approximation [17]:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_b + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u}$ is the instantaneous fluid velocity vector, p is the instantaneous fluid pressure and ρ_a is the fluid or air density. Two external forces are present in the right-hand side of Equation (1). $\mathbf{F}_b$ is the buoyancy force defined as [17]:

$$\mathbf{F}_b = -\mathbf{g} \left[\frac{T - T_o}{T_o} + 0.608(q_v - q_{vo}) - q_c \right], \quad (3)$$

Above, $\mathbf{g}$ is the acceleration due to gravity, T is the instantaneous temperature, and q_v and q_c are the instantaneous water vapor mixing ratio and liquid water mixing ratio, respectively. The subscript '0' denotes a reference value. $\mathbf{F}_c$ is an artificial external force which is applied to maintain turbulence.

The thermodynamic variables

The transport equations for instantaneous temperature T and instantaneous water vapor mixing ratio q_v are represented as [1]:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla)T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T, \quad (4)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla)q_v = -C_d + \mu_v \nabla^2 q_v. \quad (5)$$

Here, L_h is the latent heat of condensation or evaporation, c_p is the specific heat of air at constant pressure, μ_T and μ_v are the molecular diffusivities for temperature and water vapor, respectively. The term C_d indicates the rate of mass exchange between liquid water and water vapor.

The kappa-Kohler theory

The growth of particles depends on the competition between the curvature or Kelvin effect, and the solute effect or Raoult's law [18]. A spherical cloud droplet has a surface that is curved, and it consists of numerous water molecules. The greater the curvature, the greater the chance that water molecules can escape from the surface. This is curvature effect which translates into more evaporation compared to a flat surface. When solutes or dry aerosols are present, they occupy sites on the water surface that would otherwise be occupied by water molecules. This is solute effect which prevents evaporation. Combining these two effects, the Kohler theory [18] prescribes the particle equilibrium saturation ratio as [8]:

$$S = \frac{r^3 - r_d^3}{r^3 - r_d^3(1 - \kappa)} \exp \left(\frac{2\sigma_l M_l}{RT \rho_l r} \right). \quad (6)$$

Here, S is the particle equilibrium saturation ratio, r_d is the dry radius, r is the wet radius, M_l is the molar mass of water, ρ_l is the density of water, σ_l is the surface tension of water, κ is the hygroscopicity parameter of the solute, R is the universal gas constant, and T is the mean temperature. Then, we can write-

$$S = 1 + \frac{A}{r} - \kappa \frac{r_d^3}{r^3}. \quad (7)$$

where

$$A = \frac{2\sigma_l M_l}{RT \rho_l}, \quad (8)$$

The particle equilibrium supersaturation S_k can be found as

$$S_k = S - 1. \quad (9)$$

Using Equation (7), if we plot S against r , we shall get the Kohler curve [19]. The critical equilibrium saturation ratio and the critical radius can be found at the maximum of the Kohler curve as shown in Equation (10).

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}}, \quad S_c = \frac{2}{\sqrt{\kappa}} \left(\frac{A}{3r_d} \right)^{\frac{3}{2}}. \quad (10)$$

For an aerosol particle to be activated into a cloud droplet, it needs to grow beyond its critical radius. Similarly, for a cloud droplet to be deactivated into an aerosol particle, it needs to shrink below the critical radius.

Droplet growth equations

We represent the motion of each droplet based on the Lagrangian description. The equations for droplet position and droplet velocity can be written as [4]:

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad (11)$$

$$\frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}. \quad (12)$$

Above, subscript 'i' denote individual cloud droplet, $\mathbf{V}$ is the droplet velocity vector, and $\mathbf{X}$ is the droplet position vector. The particle response time is τ_p which can be written as [4]

$$(\tau_p)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu}, \quad (13)$$

where ν is the kinematic viscosity of air, ρ_l is the density of liquid water that the cloud droplets are made of, and r_i is the radius of the i -th droplet. Note that Equation (13) is valid for Stokes particles when the Reynolds number based on the relative velocity between particle and bulk flow is significantly less than 1. $\mathbf{u}_i$ is the fluid velocity at i -th particle position and is obtained through a trilinear interpolation of the Eulerian field. Also, $\mathbf{g}$ is neglected when sedimentation (falling of cloud droplets relative to air) is neglected. The change in particle radius during condensation (or evaporation) can be written as [20]

$$\frac{dr_i(t)}{dt} = \frac{G}{r_i(t)} (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (14)$$

where S_e is the environmental supersaturation from the Eulerian field and then interpolated to the particle position.

$$S_e = \frac{q_v}{q_{v,s}} - 1, \quad (15)$$

Again, $q_{v,s}$ is the saturation water vapor mixing ratio and can be calculated as [21]:

$$q_{v,s} = 621.97 \frac{p_{sat}}{p - p_{sat}}, \quad (16)$$

p_{sat} is the saturation water vapor pressure and can be calculated as [22]:

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right), \quad (17)$$

In Equation (9), G is the growth parameter and can be found as [20]:

$$G = \frac{1}{\frac{L_h \rho_l}{\mu_r T} \left(\frac{L_h}{R_v T} - 1 \right) (1 + S_k) + \frac{\rho_l R_v T}{\mu_v p_{sat}}}, \quad (18)$$

where R_v is the specific gas constant for water vapor. The change in mass during condensation (or evaporation) can be written as:

$$\frac{d(m_l(\mathbf{X}, t))}{dt} = 4\pi\rho_l \sum_{i=1}^n r_i^2(t) \frac{dr_i(t)}{dt} = 4\pi\rho_l G \sum_{i=1}^n r_i(t) (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (19)$$

where m_l is the mass of liquid water per grid cell. n is the number of cloud droplets inside each grid cell. C_d is determined as follows:

$$C_d(\mathbf{X}, t) = \frac{d(m_l(\mathbf{X}, t))}{m_a dt} = \frac{4\pi\rho_l G}{\rho_a a^3} \sum_{i=1}^n r_i(t) (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (20)$$

where m_a is the mass of air in each grid cell that has a volume of a^3 . Note that when the curvature and solute effects are not considered, $S_k = 0$.

External forcing method

The velocity field is initialized according to the method of Rogallo et al. [7] so that it demonstrates homogeneous, isotropic turbulence. But this turbulence will decay unless an external forcing is applied to maintain it. The external force can be formulated in the Fourier space as [23]:

$$\mathbf{f}_e(\mathbf{k}, t) = \epsilon_{in} \frac{\mathbf{u}(\mathbf{k}, t)}{\sum_{\mathbf{k}_f \in k} |\mathbf{u}(\mathbf{k}_f, t)|^2} \delta_{\mathbf{k}, \mathbf{k}_f}. \quad (21)$$

Above, $\mathbf{u}(\mathbf{k}, t)$ is the Fourier-transformed velocity field, $\mathbf{k}_f$ is chosen from a wavenumber shell. We try two ranges- $1 \leq k \leq 3.906$, and $1 \leq k \leq 17.378$. ϵ_{in} is the input energy dissipation rate and $\delta_{\mathbf{k}, \mathbf{k}_f}$ is a delta function. Turbulence intensity can be varied by changing the magnitude of forcing. The dissipation rate ϵ is calculated as [24]

$$\epsilon = \nu \left(\frac{\partial u'_i}{\partial x_j} \frac{\partial u'_i}{\partial x_j} \right). \quad (22)$$

The Kolmogorov length scale is

$$\eta = \left(\frac{\nu^3}{\epsilon} \right)^{1/4}. \quad (23)$$

Note that dissipation rate is dynamically updated based on Equation (22). In this work, those three wavenumber ranges correspond to low, high, and highest turbulence intensities, respectively. For lowest turbulence intensity, no forcing is involved, and turbulence is allowed to decay.

IV. Numerical Implementation

We have used the FrontTier CFD package, developed by the Applied Mathematics and Statistics Department at Stony Brook University [25], to carry out the PR-DNS calculations. The convective term is solved with the fifth order Weighted Essentially Non-Oscillatory (WENO) scheme. The viscous diffusion term is solved using the Crank-Nicolson scheme. A projection scheme is used, and the Poisson equation is solved to obtain pressure. The PETSc package [26] is utilized for the iterative linear solvers. The scalar field equations in (4) and (5) are also solved using the combination of fifth order WENO and Crank-Nicolson schemes. The equations for the droplets in (11) and (12) are solved using the Implicit Euler method. Parallelization is achieved by the domain decomposition and the MPI protocol.

V. Computational Setup

The computational domain is set to be 0.512^3 m^3 . The boundary conditions are triply periodic. The initial cloud fraction is 0.5 and a slab-like cloud configuration is assumed [1]. We conduct two studies- Study I and Study II. The first is for condensation and activation of aerosol particles into cloud droplets and the second is for evaporation and deactivation of cloud droplets into aerosol particles.

In Study I, the cloudy region is supersaturated [4], and the clear-air region is just saturated. The initial vapor mixing ratio field is specified as

$$q_v(z, t_0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & \text{elsewhere} \end{cases} \quad (24)$$

Above, $q_{v,s} = 3.872549 \text{ g/kg}$, $q_v^{max} = 1.02 \times q_{v,s} = 3.949 \text{ g/kg}$, and $q_{v,e} = 3.872549 \text{ g/kg}$. Also, $d = L/2$ is the width of the cloud slab. By imposing a neutral buoyancy condition or putting $\mathbf{F}_b = 0$ and $q_c = 0$ in Equation (3), the temperature field is initialized as [4]

$$T(t = 0) = T_o - 0.608 T_o (q_v(t = 0) - q_{vo}). \quad (25)$$

where, T_o and q_{vo} are the ensemble averages.

In the beginning, a total of 8.8×10^6 aerosol particles, are randomly placed in the cloudy area giving a number concentration of 65.565 cm^{-3} . We simulate four cases- A, B, C, and D. The differences amongst them are provided in Table 01. The mesh size is 64^3 for all cases. The initial size of all particles is $0.1 \mu\text{m}$ in case A. Our simulations start from the dry aerosol sizes in cases B to D. For a monodisperse distribution, the dry aerosol size (radius) is $0.1 \mu\text{m}$. For a lognormal distribution, the geometric mean is $0.1 \mu\text{m}$ and the standard deviation is 0.15. Level of turbulence intensity can be high or low.

Table 01: Test cases

Case	Dry Aerosol Size Distribution	Turbulence Intensity	Curvature and Solute Effects
Case A	N/A	Low	No
Case B	Monodisperse	Low	Yes
Case C	Lognormal	Low	Yes
Case D	Lognormal	High	Yes

In Study II, the cloudy region is supersaturated like Study I. However, the environment or clear-air region is subsaturated and $q_{v,e} = 0.03138 \text{ g/kg}$. A total of 8.8×10^6 cloud droplets, each having a radius of $15 \mu\text{m}$, are randomly placed in the cloudy area at the beginning. We again simulate four cases with the same dry aerosol size distributions and turbulence intensities as Study I. The kappa value is 0.61 indicating ammonium sulfate composition.

VI. Results and Discussion

In this section, we discuss the curvature and solute effects on various microphysical processes namely condensation and evaporation. We also present an analysis of atmospheric turbulence in our PR-DNS model.

Curvature and solute effects on activation and condensation

The evolution of mean temperature, and mean supersaturation are plotted in Fig. 1(a)-(b). These plots correspond to cases A, B, and C in Study I. We observe that the rate of increase or decrease is fastest in case C compared to cases A and B. We recall that no aerosol is present in case A while the dry aerosol size distribution is monodisperse in case B but lognormal in case C. The differences in particle size are considered in the source terms of the transport equations for temperature and water vapor mixing ratio. When particles grow from smaller sizes, they experience rapid changes which are more pronounced for case C.

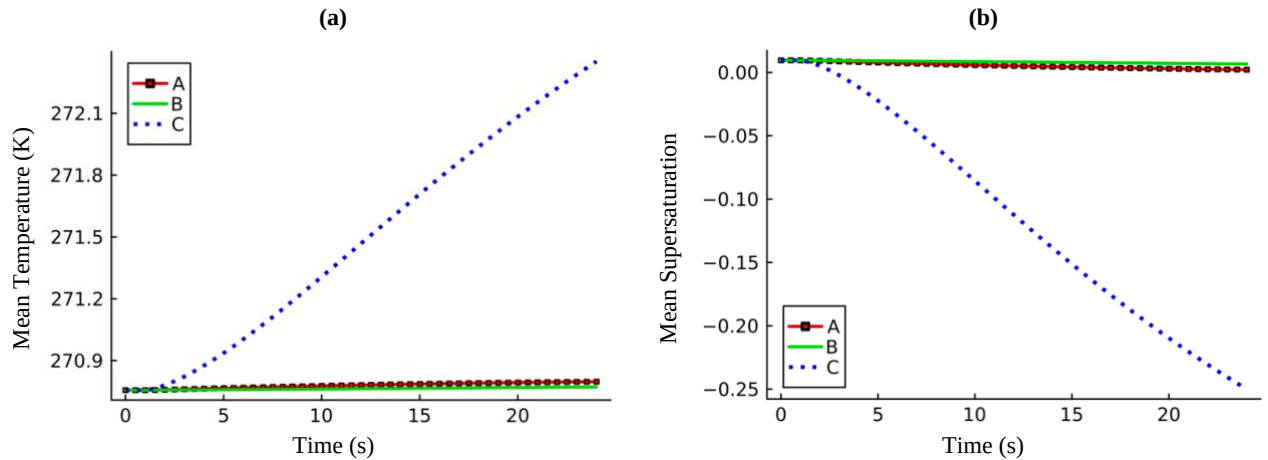


FIG. 1 Evolution of (a) mean temperature, and (b) mean supersaturation.

In Fig. 2(a)-(b), the evolution of activated aerosol number, and their mean radius are plotted. We notice that aerosol particles activate into cloud droplets over a period (3 s to 8 s) rather than at a moment in case B. Although they are of same size and exposed to the same supersaturation initially, turbulent mixing causes the supersaturation values at particles positions to be different which leads to varied particle growth. In contrast, aerosol particles grow with time but remain unactivated throughout in case A since the curvature and solute effects are ignored. However, almost half of the aerosol particles do not activate into cloud droplets in case C. We suggest that the smaller aerosol particles do not grow enough to be activated in a lognormal distribution, but the larger ones activate earlier than in case B.

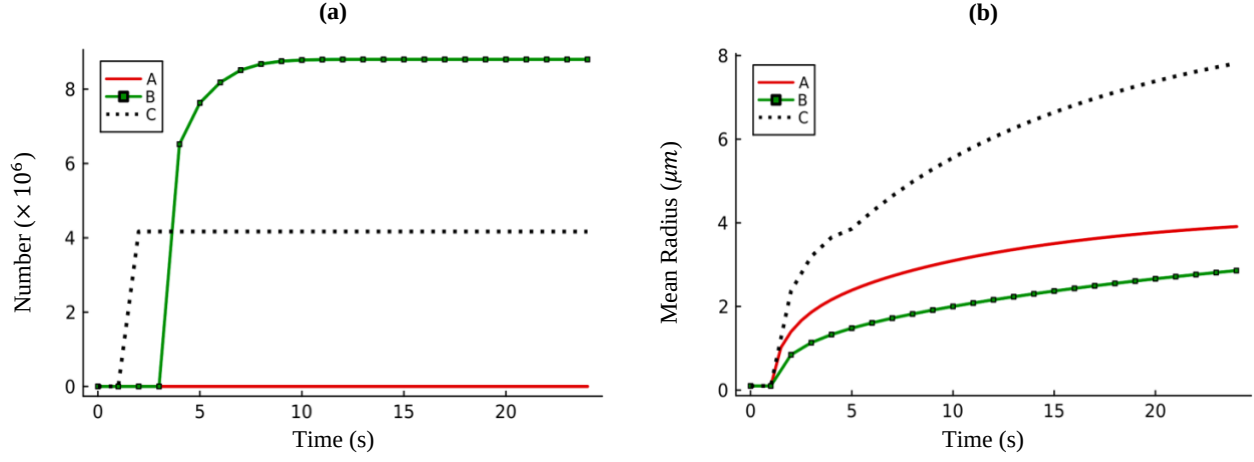


FIG. 2 Evolution of (a) number of activated aerosol particles, and (b) their mean radius.

Fig. 2(b) shows that mean radius is higher in case A than in case B. Hence, with the curvature and solute effects, particles are smaller. It is also noted that mean radius is higher in case C than in cases A and B. We reason that the larger particles in case C continue to grow bigger.

Curvature and solute effects on deactivation and evaporation

In Study II, the environment is subsaturated and cloud droplets deactivate and evaporate.

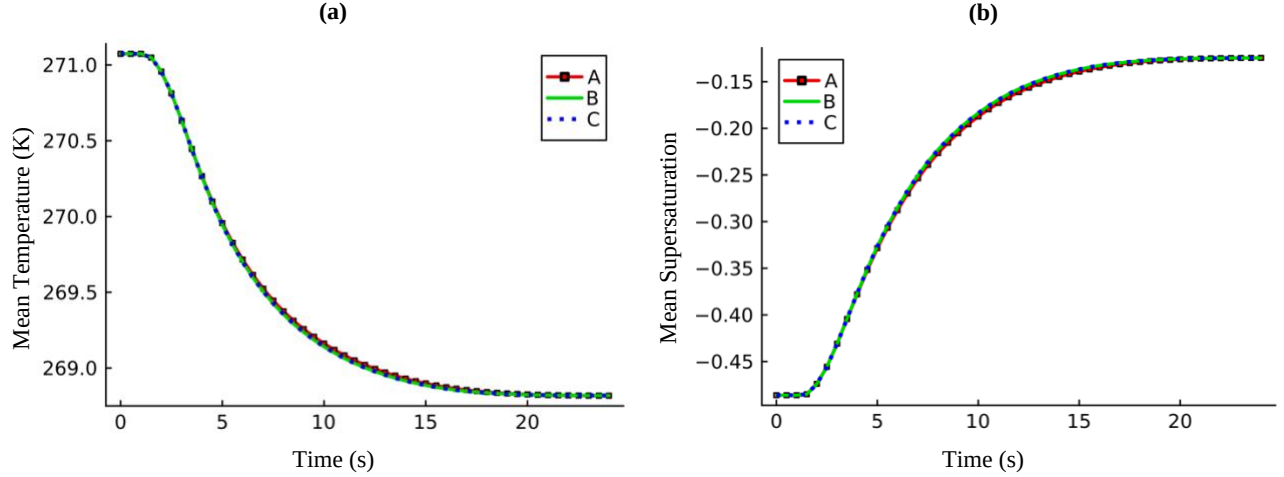


FIG. 6 Evolution of (a) mean temperature, and (b) mean supersaturation.

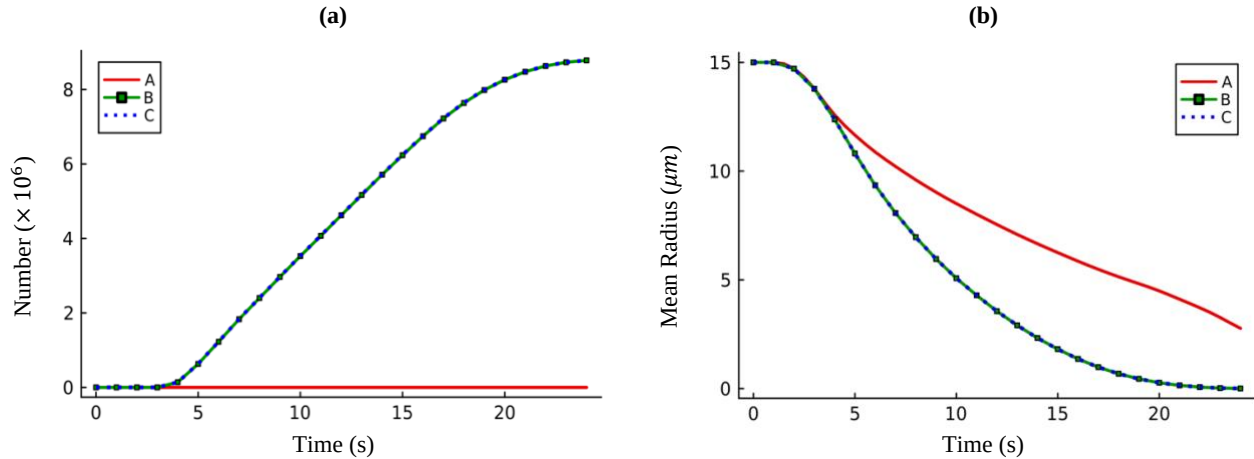


FIG. 7 Evolution of (a) number of deactivated cloud droplets, and (b) their mean radius.

Fig. 6(a)-(b) plots the evolution of mean temperature, and mean supersaturation and Fig. 7(a)-(b) plots the evolution of deactivated cloud droplet number and their mean radius. The differences amongst cases A, B, and C are negligible in Fig. 6 compared to Study I. This is due to the initial particle size being much larger in Study II.

Analysis of turbulence in our model

Turbulence plays an important role in the climate models and is therefore of interest in this work. The turbulent kinetic energy (TKE) can be estimated as

$$TKE = \frac{1}{2} (\overline{(u')^2} + \overline{(v')^2} + \overline{(w')^2}), \quad (26)$$

where, u' , v' and w' denote the fluctuating velocity components in the x-, y- and z-coordinate directions, respectively. The energy spectrum of turbulence $E(k)$ is related to the physical-space mean TKE per unit mass as

$$TKE = \int_0^\infty E(k) dk. \quad (27)$$

In Fig. 11, we compare the TKEs and dissipation rates for two levels of turbulence intensity. The TKE values in Fig. 11(a) becomes steady after first few seconds. We see in Fig. 11(b) that steady dissipation rates fall between 1×10^{-3} to $3 \times 10^{-4} m^2/s^3$. The Kolmogorov scales at equilibrium range from 0.0011 m to 0.0023 m for respective turbulence intensities.

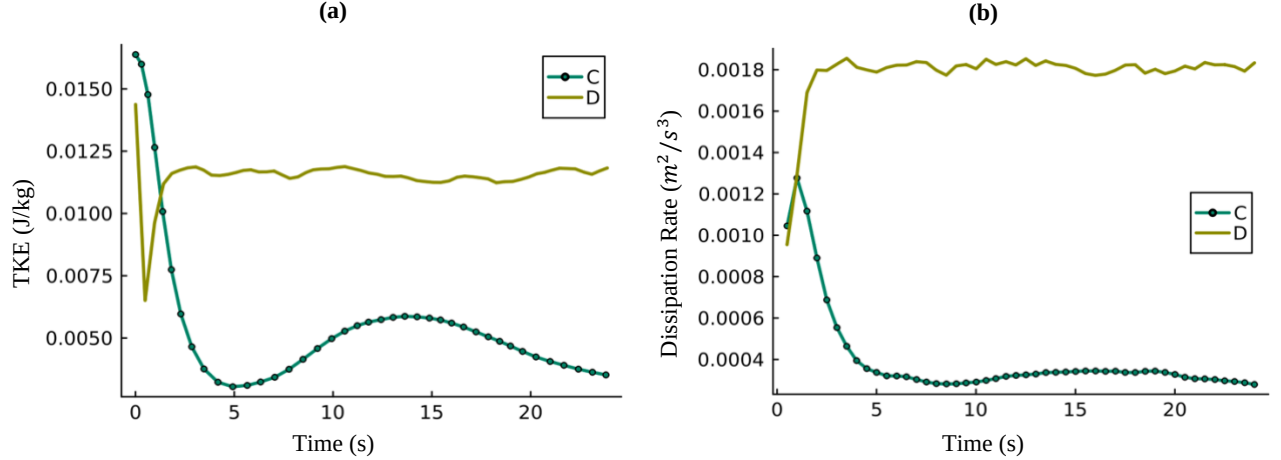


FIG. 11 Evolution of (a) TKE, and (b) dissipation rate, for two turbulence intensities.

The energy spectra at early times are shown in Fig. 12 as functions of the wavenumber. These plots are for case D or highest turbulence intensity. The kinetic energy of turbulence is mostly associated with low wavenumbers or large eddies.

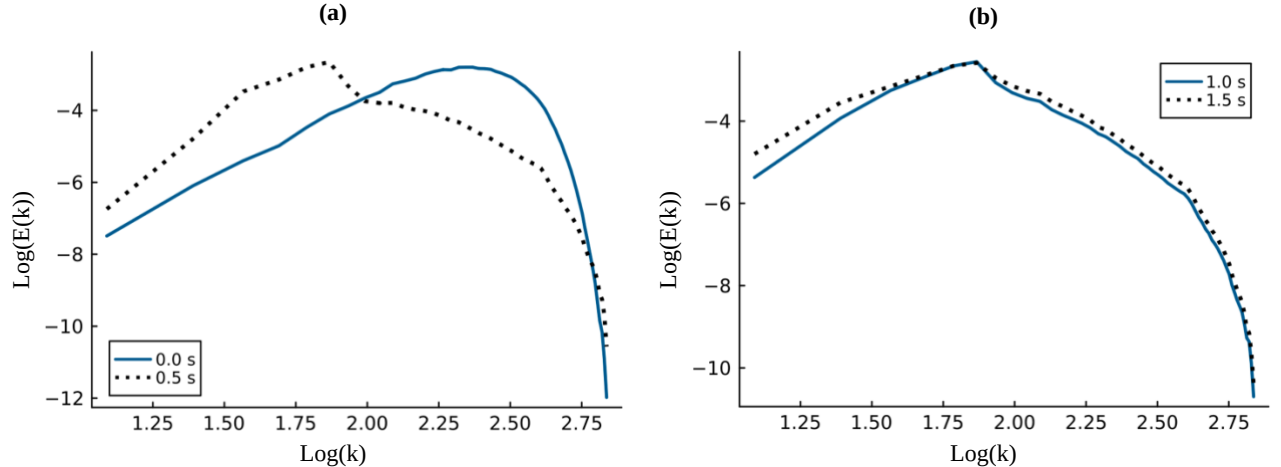


FIG. 12 log-log plots of TKE spectrum at selected times, for case D.

We define the correlation coefficients as [32]:

$$R_{ij}(\mathbf{r}) = \frac{\langle u'_i(\mathbf{x})u'_j(\mathbf{x} + \mathbf{r}) \rangle}{\sqrt{\langle (u'_i(\mathbf{x}))^2 \rangle \langle (u'_j(\mathbf{x}))^2 \rangle}}. \quad (28)$$

where, u'_i and u'_j are the fluctuating velocities, $\langle \rangle$ denotes the average, $\mathbf{x}$ is the position vector, and $\mathbf{r}$ is the separation distance vector. $i = 1, 2, 3$ represents x, y, and z coordinate directions, respectively and the same for j . If $i = j$, they are auto-correlation coefficients. If Δx , Δy , and Δz are grid spacings in the respective directions, $\mathbf{r}$ and its magnitude $|\mathbf{r}|$ can be found as

$$\mathbf{r} = \Delta x \mathbf{i} + \Delta y \mathbf{j} + \Delta z \mathbf{k}, \quad (29)$$

$$|\mathbf{r}| = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}. \quad (30)$$

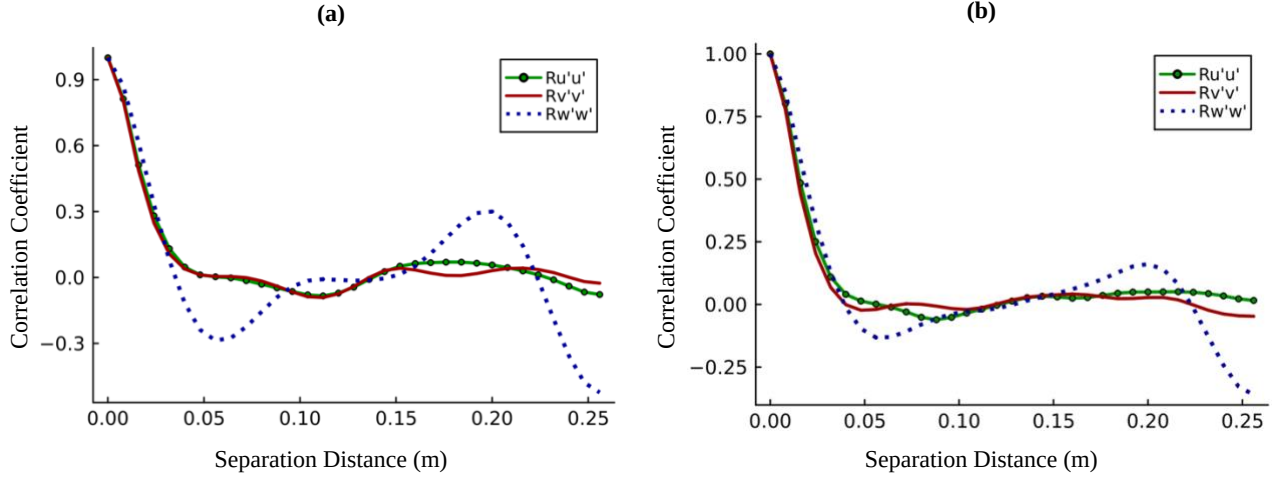


FIG. 13 Auto-correlation coefficients at (a) 1.0 s, and (d) 2.0 s, for case D.

The plots in Fig. 13 demonstrate that correlation coefficients for the fluctuating velocities approach zero as separation distance increases.

Impact of turbulence intensity

We examine the role of turbulence on activation and deactivation by comparing cases C and D. Let us be reminded that they represent low and high turbulence intensities, respectively, with dry size distribution being lognormal. We report that the mean radius of deactivated cloud droplets is larger in case D at 5 s but smaller at 16 s. This would suggest that turbulence promotes particle growth at early times but suppresses it at late times. We are conducting more investigation in this regard to reach a consensus.

VII. Conclusion

In this work, we have studied the complex turbulence-cloud-aerosol interactions by adopting a particle-resolved direct numerical simulation (PR-DNS) approach. We conclude that the curvature and solute effects are significant when droplet radius is smaller with respect to dry radius. Also, the dry size (radius) distribution influences the activation of aerosol particles. More research will be performed to determine the impact of turbulence on activation and deactivation. We hope that our efforts to understand the microscale events like condensation, evaporation, activation, and deactivation will contribute to the reduction of uncertainty in climate models.

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